Weekly Report 3
Automated Biometric System for Individual Re-Identification of Mugger Crocodile

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1 Introduction

The goal of this project is to develop an automated biometric system capable of recognizing individual Mugger crocodiles using aerial images. The objective of this system is to determine whether a crocodile observed in a new image belongs to a previously known individual or represents a newly observed crocodile.

In this phase, we focused on two major improvements:

- Fine-tuning Inception-v3 for the identification task using a transfer learning approach.
- Preparing annotated data for object detection that will allow automatic extraction of the region which contains unique biometric patterns for crocodile identification.

2 Dataset Preparation

The dataset consists of aerial UAV images of Mugger crocodiles and contains **6 crocodile identity classes**, with a total of **4094 images**. The images were split into training and validation sets using a **90:10 ratio**.

- Classes: 6
- Class IDs: 107, 108, 109, 110, 111, 112
- Total images: 4094
- Training images: 3684
- Validation images: 410

Additionally, We manually annotated **300 images** using **Label-Studio**. Bounding boxes were drawn around the crocodile, which contains unique scale patterns useful for identification. All annotations were assigned the class label and will be exported in **YOLO format** for future object detection training.

Figure 1: Example annotation of the dorsal scute region used for YOLO training.

3 Baseline Model Implementation

To improve the baseline system, we fine-tuned the **Inception-v3 model** pre-trained on the ImageNet dataset and applying tranfer learning principles.

The following modifications were introduced during fine-tuning:

- The final fully connected classification layer of Inception-v3 was replaced to match the number of crocodile identities present in the dataset.
- The auxiliary classifier of the network was also modified to output the same number of classes.
- Cross-entropy loss was used as the primary training objective.

4 Experimental Setup

The experiments were implemented using the **PyTorch deep learning framework**. Training was performed using GPU acceleration when available. The following hyperparameters were used during training:

Parameter	Value
Model Architecture	Inception-v3 (Pretrained)
Task	RE-ID Classification
Optimizer	RMSprop
Learning Rate	0.01
Batch Size	16
Epochs	20
Input Image Size	299 × 299
Train/Validation Split	90:10
Loss Function	Cross-Entropy

Training performance was monitored using several evaluation metrics:

- Training Accuracy
- Validation Accuracy
- Training Loss
- True Positive Rate (TPR)
- True Negative Rate (TNR)
- False Positive Rate (FPR)

To evaluate open-set recognition capability, below thresholds was used:

$$\tau = 0.84$$

This threshold determine whether a prediction should be considered a known crocodile identity or classified as an unknown individual.

5 Preliminary Results

The fine-tuned RE-ID model demonstrated the ability to learn visual patterns associated with individual crocodiles.

During training, the model produced learning curves representing training accuracy, validation accuracy, and loss values over the training epochs.

The training process generated the following outputs:

- Training accuracy curve
- Validation accuracy curve
- Training loss curve
- Saved trained model

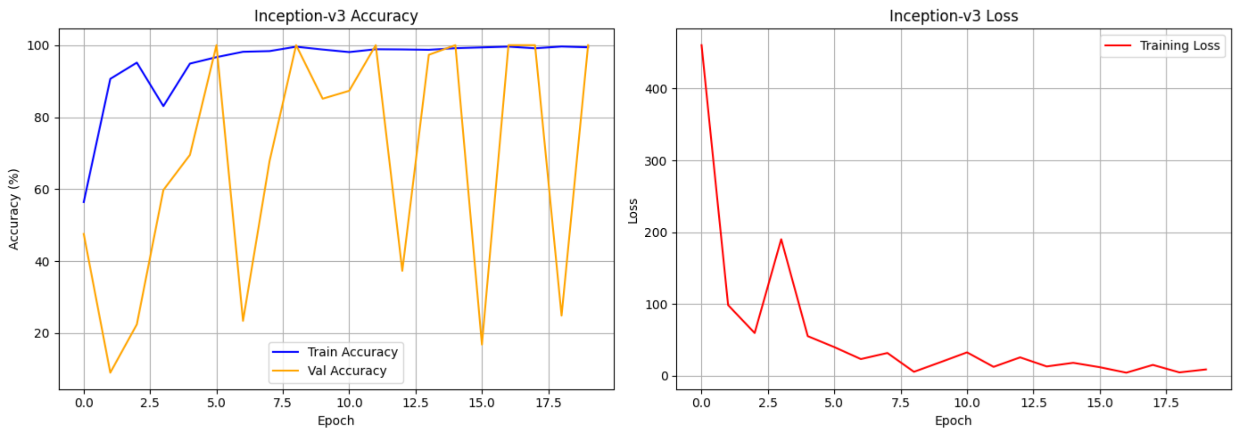

Figure 2: Training, Validation Accuracy and Training Loss across Epochs

Metric	Result
Overall Accuracy on Test Data	8.3%
True Positive Rate	16.7% (correctly re-identified known crocs)
True Negative Rate	0.0% (correctly rejected unknown crocs)
False Positive Rate	100.0% (wrongly accepted unknowns as known)

6 Model Evaluation and Visualization

To further analyze the performance of the trained Inception-v3 RE-ID model, additional evaluation plots were generated. These plots help visualize the relationship between classification thresholds and detection performance, as well as provide qualitative insight into the model predictions.

6.1 Prediction Visualization

To qualitatively evaluate the model predictions, a visualization of the test images was generated, as shown in Figure 3. Each image displays the predicted crocodile identity along with the associated confidence score.

The results show that while the model is able to confidently assign identities to images, some unknown crocodiles are incorrectly classified as known identities. This behavior highlights the challenge of open-set recognition in wildlife identification tasks.

Figure 3: Visualization of model predictions on test images. Each image shows the predicted identity and confidence score produced by the trained Inception-v3 model.

7 Challenges Faced

Several challenges were encountered during this phase of the project:

- UAV imagery often contains variations in lighting, orientation, and camera angle.
- Crocodiles may appear partially submerged in water, making feature extraction difficult.
- Background environments such as mud, vegetation, and water can introduce noise in the images.

8 Plan for Next Week

We will first focus on identifying a suitable threshold and debugging the reasons behind the low accuracy results. After resolving these issues, we will proceed with the following tasks:

YOLO Phase

- Fine-tune YOLOv5l using the 300 annotated images.
- Run the trained detector on all 4,094 images.
- Detect dorsal scute regions automatically.
- Crop the detected regions from each image.
- Store the cropped images with the original crocodile identity labels.

RE-ID Phase

- Use cropped scute images as input for the RE-ID model.
- Train the model on approximately 680 images per crocodile identity.
- Use a 90:10 training-validation split.
- Train for 20 epochs using RMSprop optimizer.

Evaluation Phase

- Evaluate the model on known crocodile identities.
- Measure True Positive Rate (TPR).
- Evaluate performance on unknown crocodiles.
- Measure True Negative Rate (TNR).
- Plot threshold curves to determine optimal decision thresholds.

These steps will lead to the development of a robust automated biometric identification system for Mugger crocodiles.